

# SEC P vs NP — Frameworks (live)

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-21 12:07 UTC

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Compiled 2026-05-21 12:07 UTC. Tracking 6 active + 0 dead frameworks.

### Summary table

| Framework ID  | Status      | Fitness | Generation | Parent | Name                                                          |
|---------------|-------------|---------|------------|--------|---------------------------------------------------------------|
| fw_b9e7d103d0 | ELABORATING | 0.000   | 0          | -      | Ergodic<br>Circuit<br>Frame-<br>work                          |
| fw_28b4bfb95f | ELABORATING | 0.000   | 0          | -      | TROPICAL_FOURIER                                              |
| fw_85a254b4a0 | ELABORATING | 0.000   | 0          | -      | Coarse<br>Geomet-<br>ric<br>Karchmer-<br>Wigderson<br>(CG-KW) |
| fw_6997a27304 | ELABORATING | 0.000   | 0          | -      | Coarse<br>Geomet-<br>ric<br>Lifting<br>(CGL)                  |
| fw_a1a152ae17 | ELABORATING | 0.000   | 0          | -      | Tropical<br>Circuit<br>Weight<br>Analysis<br>(TCWA)           |
| fw_6ef88ddbf3 | ELABORATING | 0.000   | 0          | -      | Coarse-<br>Geometric<br>KW<br>Bounds<br>(CG-KW)               |

### Details

#### Ergodic Circuit Framework (fw\_b9e7d103d0)

- **Status:** ELABORATING
- **Fitness:** 0.000

- **Taxonomy:** COMM\_COMPLEXITY
- **Target invariant:** communication\_entropy\_barrier (real number  $\geq 0$ )  $\rightarrow$  bounds The minimum communication complexity in a k-party number-on-forehead model, where lower bounds are derived from the growth rate of Kolmogorov-Sinai entropy in associated dynamical circuits. Aims to prove  $\omega(\log n)$  lower bounds for explicit functions like Disjointness, avoiding natural proofs via non-uniform dynamics.

**Primitives:** - measurable\_dynamical\_circuit (tuple  $(C, X, \mu, T)$ ): A Boolean circuit  $C$  where gates are labeled by operations over a probability space  $(X, \mu)$ , and  $T: X \rightarrow X$  is a measure-preserving transformation. - orbit\_signature (function:  $N \rightarrow [0,1]$ ): For a fixed input  $x \in \{0,1\}^n$ , the orbit signature is the sequence  $(\mu(C(T^k(x))))_k$ , capturing how the circuit's output evolves under repeated application. - cross\_correlation\_flow (matrix  $\in R^{d \times d}$ ): For a depth- $d$  circuit, this matrix  $F_{i,j} = \int |C_i(x) - C_j(T^j(x))|^2 d\mu(x)$  measures how perturbations via  $T$  propagate through layers. Computable via

**Tentative axioms:** - A1: Any circuit with  $o(n)$  communication complexity induces a dynamical system with sublinear Kolmogorov-Sinai entropy growth. - A2: Measure-preserving transformations corresponding to  $AC^0$  circuits have bounded mixing time, while those for PSPACE-complete computations exhibit super-polynomial mixer-profile decay. - A3: If a family of circuits computes a function with high cross\_correlation\_flow norm, then its communication complexity is  $\Omega(n^\delta)$  for some  $\delta > 0$ .

## TROPICAL\_FOURIER\_ANALYSIS (fw\_28b4bfb95f)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** FOURIER\_ANALYTIC
- **Target invariant:** MinimalFourierCoefficient (RealNumber)  $\rightarrow$  bounds The discrepancy of a function under tropical Fourier analysis.

**Primitives:** - TropicalPolynomial (Function): A function defined over a tropical semiring (max-plus or min-plus) with computable coefficients. - TropicalFourierTransform (Transformation): A mapping from tropical polynomials to their Fourier coefficients over a tropical semiring. - DiscrepancyMeasure (Metric): A function quantifying the deviation of a tropical polynomial from uniformity.

**Tentative axioms:** - A1: TropicalConvolution preserves the tropical semiring structure under composition. - A2: TropicalFourierTransform is invertible when restricted to certain tropical polynomials. - A3: DiscrepancyMeasure is bounded by the maximum absolute value of Fourier coefficients.

## Coarse Geometric Karchmer-Wigderson (CG-KW) (fw\_85a254b4a0)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** KARCHMER\_WIGDERSON
- **Target invariant:** Coarse depth  $\kappa(f)$  (non-negative real)  $\rightarrow$  bounds  $\kappa(f) := \sup$  over nontrivial  $c \in HX^1(X_f)$  and  $T \in C_u^*[X_f]$  of  $\log|\langle c, T \rangle| / \log(\text{propagation}(T))$ . The conjecture is  $\text{depth}(f) \geq \kappa(f)$ , so a super-logarithmic  $\kappa(f)$  for an explicit  $f$  gives a super-logarithmic formula-depth lower bound (a  $P \not\subseteq NC^1$  separation if  $\kappa$  is poly-large).

**Primitives:** - KW-metric space (finite metric space  $(X_f, d_f)$  with  $X_f = f^{-1}\{0\} \sqcup f^{-1}\{1\}$ ): For a boolean function  $f: \{0,1\}^n \rightarrow \{0,1\}$ , define  $d_f(x,y) = \log_2(\text{min number of coordinates a depth-bounded distinguisher must read to certify } f(x) \neq f(y))$ . - Controlled cover (finite family  $U = \{U_i \subset X_f\}$  with diameter bound  $R$ ): A cover whose parts have  $d_f$ -diameter at most  $R$  and whose overlap multiplicity is at most  $m$ . Encoded as a hypergraph  $H_U$  (vertices =  $X_f$ , edges =  $U_i$ ). - Coarse cocycle (function  $c: X_f \times X_f \rightarrow \mathbb{Z}$  with finite support modulo controlled equivalence): A discrete 1-cochain in the Roe-style coarse cohomology  $HX^1(X_f)$  restricted to the KW pair structure. Stored

as a sparse table on pairs  $(x,y)$  with  $d_f$  - Asymptotic-dimension witness (vertex coloring  $\chi : X_f \rightarrow [k]$  together with diameter bound  $R$ ): Witnesses  $\text{asdim}(X_f) \leq k-1$  in the sense of Gromov: each color class decomposes into  $d_f$ -bounded pieces of diameter  $\leq R$  that are pairwise  $R$ -disjoint.  $C$  - Roe-controlled operator (matrix  $T \in R^{\{X_f \times X_f\}}$  with  $T_{\{x,y\}} = 0$  whenever  $d_f(x,y) > R$ ): Element of the algebraic uniform Roe algebra  $C_u^*[X_f]$ . Stored as a banded sparse matrix in the  $d_f$ -metric; multiplication, adjoint, and trace are al

**Tentative axioms:** - A1 (Coarse-KW link): formula depth  $d(f) \geq \Omega(\log(\text{asdim}(X_f)))$  and more strongly  $d(f) \geq \Omega(\kappa(f))$ ; proved by translating a KW protocol into a controlled cover whose multiplicity exponent equals depth. - A2 (Composition sub-additivity):  $\text{asdim}(X_{\{f \circ g\}}) \geq \text{asdim}(X_f) + \text{asdim}(X_g) - O(1)$ , giving a KRW-style direct-sum theorem at the level of coarse dimension. - A3 (Anti-natural-proofs): for a uniformly random  $f$ ,  $HX^1(X_f)$  is generically zero (Roe-algebraic concentration of measure), so  $\kappa$  is NOT a ‘largeness’ property in the Razborov-Rudich sense — random  $f$ ’s - A4 (Anti-relativization):  $\kappa$  is a metric invariant of  $d_f$  and is destroyed by oracle access (oracle gates collapse  $d_f$  to  $\leq 1$ ), so  $\kappa$ -based bounds cannot relativize, in the spirit of Mendel-Naor metric - A5 (Roe-index rigidity): nontrivial classes in  $HX^1(X_f)$  detected by the Roe-trace pairing are stable under coarse equivalence, mirroring Yu’s coarse Baum-Connes results (Yu 2000; Nowak-Yu 2012); expl

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## Coarse Geometric Lifting (CGL) (fw\_6997a27304)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** LIFTING
- **Target invariant:** CoarseLiftingComplexity (CLC) ( $N$ -valued function of  $(f, G) \rightarrow$  bounds The deterministic 2-party communication complexity of  $f \circ G^n$ . Specifically:  $CC(f \circ G^n) \geq Q(f) \cdot \log_2(\text{asdim}_R(G)+1)$  for the gadget scale  $R = \text{diam}(G)$ , with the goal of recovering and surpassing known query-to-communication lifting bounds (Raz-McKenzie, Göös-Pitassi-Watson) for gadgets where positive asymptotic dimension can be certified.

**Primitives:** - MetricGadget ( $(X: \text{FiniteSet}, Y: \text{FiniteSet}, g: X \times Y \rightarrow \{0,1\}, d: (X \times Y)^2 \rightarrow \mathbb{R}_{\geq 0})$ ): A boolean gadget  $g$  equipped with a graph/Hamming metric  $d$  on  $X \times Y$ . Implementable as a tuple of two finite sets, a truth-table for  $g$ , and a precomputed - LiftedInputSpace  $((X \times Y)^n, d_{\oplus})$ : The  $n$ -fold product of a MetricGadget under the  $\square^1$  (Hamming-sum) product metric  $d_{\oplus}((u_i), (v_i)) = \sum d(u_i, v_i)$ . Implementable as a generator over tup - RoeCover ( $\text{List}[\text{Subset}((X \times Y)^n)]$  with multiplicity  $m$  and scale  $R$ ): A finite cover of the lifted input space whose members have  $d_{\oplus}$ -diameter  $\leq R$  and any point is in  $\leq m$  members. This is the discrete witness used in the - ProtocolPullback ( $\text{Dict}[\text{transcript} \rightarrow \text{Subset}((X \times Y)^n)]$ ): The partition of inputs induced by transcripts of a (deterministic or randomized) communication protocol  $\Pi$ , viewed as a candidate RoeCover. Implementa - FolnerWitness ( $\text{Sequence}[\text{Subset}((X \times Y)^n)]$ ): A nested family  $F_1 \subset F_2 \subset \dots$  with  $|\partial_R F_k|/|F_k| \rightarrow 0$  for each scale  $R$ , certifying amenability/Property A of the lifted space relative to a gadget.

**Tentative axioms:** - A1 (Coarse-Lift Functoriality): If two gadgets  $G_1, G_2$  are coarsely equivalent (a bi-Lipschitz bijection up to bounded error), then  $\text{CLC}(f, G_1) = \Theta(\text{CLC}(f, G_2))$  for every  $f$ . - A2 (Protocol  $\rightarrow$  Cover): Every deterministic protocol  $\Pi$  for  $f \circ G^n$  of cost  $c$  yields a RoeCover with multiplicity  $\leq 2^c$  and bounded scale  $R_{\Pi}$  depending only on  $G$ ; hence  $CC(f \circ G^n) \geq \log_2(\min \text{multiplicity})$  - A3 (Asdim Amplification):  $\text{asdim}(G_1 \otimes G_2) \geq \text{asdim}(G_1) + \text{asdim}(G_2) - O(1)$ , so iterated tensoring of a single ‘good’ gadget produces unbounded coarse dimension. - A4 (Property-A Transfer): If  $G$  has Property A and  $f$  has query complexity  $Q(f)$ , then  $\text{CLC}(f, G) \geq Q(f) \cdot h(G)$  where  $h(G)$  is the Hilbert-space compression exponent of  $G$  — a non-natural, non-relativizing - A5 (Non-Algebrization): The Roe algebra  $C^*_R((X \times Y)^n)$  is not closed under polynomial extensions of the gadget alphabet; therefore lower bounds derived from its  $K$ -theory do not algebrize.

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## Tropical Circuit Weight Analysis (TCWA) (fw\_a1a152ae17)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** BOUNDED\_ARITHMETIC
- **Target invariant:** Tropical Proof Rank (integer)  $\rightarrow$  bounds The minimum number of phase transitions required to simulate a bounded arithmetic proof line in  $V^0$  or  $\text{IA}_0$ , thereby bounding the strength of definable functions and aiming to separate bounded arithmetic theories via circuit phase complexity.

**Primitives:** - Tropical Circuit (tuple  $(G, w)$ ): A directed acyclic graph  $G$  with nodes labeled as inputs (variables or constants), tropical addition (min or max), or tropical multiplication (usual ad  
- Tropical Derivation (function  $\delta: \text{Circuit} \rightarrow \text{Circuit}$ ): A formal derivative operator on tropical circuits: for a node computing  $\min(x, y)$ , the derivation selects the active operand (argmin) with tie-breaking - Weight Profile (list of integers): For a tropical circuit  $C$  and input  $x \in \mathbb{Z}^n$ , the weight profile is the sequence of edge weights traversed during evaluation under tropical semantics, or  
- Circuit Phase Space (subset of  $\mathbb{R}^n$ ): Partition of input space  $\mathbb{R}^n$  into convex polyhedral regions where the tropical circuit's active computation path (i.e., argmin/argmax decisions) remain

**Tentative axioms:** - A1: (Phase Stability Axiom) Any function provably total in  $\text{IA}_0$  has a tropical circuit with polynomially many phase cells under uniform input scaling. - A2: (Weight-Proof Correspondence) The sum of absolute weights in a tropical circuit for a formula encoding a proof line is at least the bit-complexity of the cut-free proof in bounded arithmetic. - A3: (Tropical Soundness) If a bounded arithmetic theory proves a  $\forall \Sigma_1^b$  sentence, then its tropical circuit model exhibits a homotopy-stable phase space under perturbation.

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## Coarse-Geometric KW Bounds (CG-KW) (fw\_6ef88ddb3)

- **Status:** ELABORATING
- **Fitness:** 0.000
- **Taxonomy:** KARCHMER\_WIGDERSON
- **Target invariant:** formula\_depth\_D(f) (natural\_number)  $\rightarrow$  bounds  $\log_2$  of minimum size, equivalently minimum depth, of a De Morgan formula computing  $f$ . The framework conjectures  $D(f) \geq c * \text{asdim}(X_f)$  and  $D(f) \geq c' / \alpha * (X_f)$ , giving super-logarithmic depth lower bounds when  $X_f$  has no Hilbert compression (Yu 2000; Nowak-Yu 2012).

**Primitives:** - KW\_metric\_space (metric\_space): For boolean  $f: \{0,1\}^n \rightarrow \{0,1\}$ , the set  $X_f = f^{-1}(0) \times f^{-1}(1)$  equipped with the disagreement-Hamming metric  $d((x,y), (x',y')) = |\{i : (x_i \neq y_i) \vee (x'_i \neq y'_i)\}|$   
- AsymDim\_cover (indexed\_cover): A family  $\{U_a\}_{a \in A}$  of subsets of  $X_f$  with uniformly bounded diameter and bounded  $r$ -multiplicity (every  $r$ -ball meets  $\leq k+1$  sets). Encoded as a hy  
- Hilbert\_compression\_embedding (Lipschitz\_map): A map  $\phi: X_f \rightarrow \mathbb{R}^N$  with explicit upper Lipschitz constant  $L$  and lower compression  $\rho(t) \sim t^\alpha$ , certified by a Gram matrix and PSD check. Compu - Coarse\_partition\_tree (rooted\_labeled\_tree): A binary tree whose leaves partition  $X_f$  into 'monochromatic' coordinate-rectangles; each internal node is labeled with a coordinate  $i$  in  $[n]$  separati - Property\_A\_witness (function\_family): Maps  $\xi: X_f \rightarrow \ell_1(X_f)$  of finite support such that  $\|\xi_x - \xi_y\|_1$  is small when  $d(x,y)$  bounded, and supports lie in fixed-radius balls. Verifia

**Tentative axioms:** - A1 (Tree-to-Dimension): Every depth- $d$  KW protocol yields an AsymDim\_cover of  $X_f$  with multiplicity  $\leq d+1$  and bounded diameter; hence  $\text{asdim}(X_f) \leq D(f)$ . - A2 (Compression Lower Bound): If  $X_f$  admits a Hilbert embedding with compression exponent  $\alpha$ , then  $D(f) \geq \Omega(1/\alpha)$ ; contrapositively, expander-like  $X_f$  forces large depth (cf. Gromov's a-T-m - A3 (Coarse Composition / KRW):  $\text{asdim}(X_{f \circ g}) \geq \text{asdim}(X_f) + \text{asdim}(X_g) - O(1)$ , giving a coarse-geometric route to the Karchmer-Raz-Wigderson conjecture. - A4 (Non-Naturalness): The predicate ' $\text{asdim}(X_f) \geq k$ ' is neither large nor constructive in the Razborov-Rudich sense - random functions have bounded  $\text{asdim}$  with high probability because random metric  $s$  - A5 (Non-Relativizing / Non-Algebrizing): Coarse invariants depend on the fine combinatorial structure of  $f^{-1}(0)$  and  $f^{-1}(1)$  as embedded in the Hamming cube, not on oracle access or low-degree alg